(ii) $3x^2 - 2x + 1 = 0$ 16. $3, \frac{9}{4}$ 17.(i) $\begin{pmatrix} 750 & 1500 & 2250 \\ 3750 & 2250 & 750 \end{pmatrix}$ (ii) $\begin{pmatrix} 8000 & 16000 & 24000 \\ 40000 & 24000 & 8000 \end{pmatrix}$

18. $\sin\theta$ 19. 8, 4 20. $\begin{pmatrix} 122 & 71 \\ -58 & -34 \end{pmatrix}$

## Exercise 4.1

1.(i) Not similar (ii) Similar,2.5 2. 3.3 m 3. 42 m

5. $\frac{15}{13}, \frac{36}{13}$ 6. 5.6 cm, 3.25 cm 8. 2.8 cm 9. 2 m

## Exercise 4.2

1.(i) 6.43 cm (ii) 1 2. 60 cm 5. 4 cm, 4 cm

8.(i) Not a bisector (ii) Bisector 12. 2.1 cm

## Exercise 4.3

1. 30 m 2. 1 mile 3. 21.74 m 4. 12 cm, 5 cm

5. 10 m, 24 m, 26 m 6. 0.8 m

## Exercise 4.4

1. 7 cm 2. 2 cm 3. 7 cm, 5 cm, 3 cm 4. 30° 5. 130°

6. $\frac{20}{3}$ cm 7. 10 cm 8. 4.8 cm 10. 2 cm 13. 8.7 cm 14. 10.3 cm 15. 4 cm 16. 6.3 cm

## Exercise 4.5

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (C) | (B) | (D) | (A) | (D) | (A) | (B) | (C) | (A) | (D) | (B) | (B) | (B) | (D) | (A) |

## Unit exercise-4

2. $\frac{12}{5}$ cm, $\frac{10}{3}$ cm 5. $20\sqrt{13}$ km 7. 10 m 8. shadow $= \frac{4}{11} \times$ (distance) 10. 6 units

## Exercise 5.1

1.(i) 24 sq. units (ii) 11.5 sq. units 2.(i) collinear (ii) collinear

3.(i) 44 (ii) 13 4.(i) 0 (ii) $\frac{1}{2}$ or $-1$ 5.(i) 35 sq. units (ii) 34 sq. units

6. $-5$ 7. $2, -1$ 8. 24 sq. units, $\text{area}(\Delta ABC) = 4 \times \text{area}(\Delta PQR)$ 9. 122 sq.units

10. 10 cans 11.(i) 3.75 sq. units (ii) 3 sq. units (iii) 13.88 sq. units

## Exercise 5.2

1.(i) undefined (ii) 0 2.(i) 0° (ii) 45° 3.(i) $\frac{1}{\sqrt{5}}$ (ii) $-\cot\theta$

4. 3 6. 7 7. $\frac{17}{2}$ 8. 4 9.(i) yes (ii) yes 11. 5, 2

## Exercise 5.3

1.(i) $2y + 3 = 0$ (ii) $2x - 5 = 0$ 2. $1, 45°, \frac{5}{2}$

3. $x - \sqrt{3y} - 3\sqrt{3} = 0$ 4. $\frac{\sqrt{3}+3}{2}, \frac{3+3\sqrt{3}}{-2}$ 5. $-5$ 6. $x - y - 16 = 0$

7.(i) $16x - 15y - 22 = 0$ (ii) $4x - 9y + 19 = 0$ 8. $15x - 11y + 46 = 0$

9. $x + 4y - 14 = 0$, $3x + 5y - 28 = 0$ 10. $5x + 4y - 3 = 0$

11. (i) 1 (ii) 7.5 seconds (iii) 10 seconds



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